

A Reproducible, Pre-Registered Test of Mechanical Elliott-Wave Signals, and Whether a Local-LLM Scenario Weigher Improves Calibration

Thomas · Elliott Wave RCT project
Working paper — deterministic and local-LLM arms complete, with statistics

Abstract

We build a deterministic, point-in-time Elliott-Wave (EW) signal generator and an independent reliability tester sharing one set of outcome rules, and ask (1) whether the mechanical engine yields a tradeable, calibrated edge, and (2) whether a local LLM scenario weigher improves it. On an anonymized 50-instrument panel (1962–2026; 588k daily bars) the engine produced 68 distinct out-of-sample setups with a 26.5% win rate (Wilson 95% CI 17.4–38.0%) and +0.35R raw expectancy (bootstrap 95% CI -0.10 to $+0.83$; $p=0.15$ — not significant). The result is market drift, not wave skill: all profit comes from long setups (+0.89R; the 22 short setups lose, $-0.79R$), simply going long on every setup beats the engine (+0.63R), and against a buy-and-hold benchmark the engine's alpha is significantly negative ($-6.16R$, 95% CI -7.97 to -4.46). Two further controls agree: tuning the engine's most powerful parameter created an edge that vanished once both the parameter and the instruments were held out ($p=0.59$), and a gradient-boosted weigher's apparent skill was entirely the direction sign (drift), with wave-structure features at chance (AUC 0.53). The local-LLM weigher did not help — expectancy edged down and grade calibration worsened. We report a reproducible null: no detectable wave-counting edge, and a mechanization that underperforms buy-and-hold.

1. Introduction

EW analysis is popular but hard to automate objectively; critics cite its subjectivity and the ease of fitting counts in hindsight [1,2]. We sidestep the 'one true count' by emitting ranked candidate counts with confidence, and we ask the only question that matters for use: are the signals reliable out-of-sample, is the confidence calibrated, and does the engine beat the trivial benchmark of simply holding the market? A strict generator/tester split keeps the measured system from measuring itself.

2. Related work — how prior work ranks EW

The academic mainstream is skeptical of technical analysis in general and EW in particular. Lo, Mamaysky & Wang [3] built the reference automated framework — kernel-regression pattern recognition on U.S. stocks 1962–1996, comparing conditional vs. unconditional return distributions — and found only modest incremental information in some patterns. Sullivan, Timmermann & White [4], applying White's Reality Check bootstrap to ~100 years of the DJIA (extending Brock, Lakonishok & LeBaron [5]), found that once data-snooping across the rule universe is corrected, out-of-sample profitability largely vanishes. EW-specific studies are fewer and mixed: D'Angelo & Grimaldi [6] report accurate EUR/USD forecasts 2009–2015, and regional studies (an Indian-market verification [7]; a metals cycle study [8]) claim support — but these are largely in-sample, single-market, or analyst-discretion fits. The recurring critique is the same two problems: subjectivity of the count and data-snooping / hindsight.

What we replicate. Our multi-decade, multi-stock U.S. panel (1962–2026) overlaps Lo et al.'s era and design philosophy (automate the subjective; test broadly), and we adopt the out-of-sample + bootstrap discipline of [4]. We attack the two standard critiques directly: subjectivity → a fully deterministic, reproducible engine (golden-record hashed); data-snooping/hindsight → anonymized instruments (no names/dates/prices reach the analyst or the LLM), a hard no-lookahead boundary, and pre-registration of the LLM arm.

What we do not (yet) replicate. We test tradeable setups, not Lo et al.'s conditional-return-distribution estimand, so results are not directly comparable to [3]. We now bound data-snooping across the pivot-sensitivity parameter with a disjoint-selection test (§5.1) but do not yet run White's Reality Check / Hansen's SPA across a full rule universe [4]. And unlike the positive regional EW studies [6–8], we forbid in-sample fitting and analyst-chosen counts — which plausibly explains why our out-of-sample result is weaker than their in-sample claims.

3. Methods

3.1 Data. 50 anonymized instruments (stock_id only), 588,427 daily OHLCV bars, 1962–2026, positive prices, no gaps.

3.2 Engine. Log-aware ZigZag pivots → bounded, beam-pruned, bidirectional sweep over motive (impulse, diagonals) and corrective (zigzag, flat, triangle) structures → scale-aware cardinal rules ($W2 < W1$; $W3$ not shortest; $W4$ overlap) + guideline scores (Fibonacci bands, alternation, equality, volume) → three-degree nesting (daily $\subset$ weekly $\subset$ monthly) → weighted scenarios + residual bucket → R/R-gated setup. Large moves are judged on log scale.

3.3 Setup construction. Entry/stop anchored to the completion pivot; reward/risk on log scale; degree-scaled invalidation and stop; degree-scaled evaluation horizon; stable ids to de-duplicate one structure re-seen across bars.

3.4 No-lookahead & determinism. A signal at T uses only bars $\leq T$ (as_of clamp + tripwire test); deterministic, golden-record hashed; frozen JSON carries data hash + engine version.

3.5 Tester & statistics. An independent scorer resolves each setup against the continuation (won/lost/invalidated/expired; stop-first tie-break) and reports, per arm: win rate with a Wilson 95% interval; expectancy in R with a bootstrap 95% interval and a two-sided test that $\text{mean} \neq 0$; a random-direction permutation null (each setup's direction flipped by a coin, geometry mirrored in log space) for directional skill; a drift benchmark (buy-and-hold from entry, and every setup forced long) to separate market drift from wave skill; and calibration by grade and confidence bucket.

4. Results — deterministic engine

2,306 point-in-time signal records across the 50 instruments (annual walk-forward, full history) yielded 68 distinct setups (the engine declines the great majority of the time).

Metric	Value	Inference
Win rate	26.5%	Wilson95 [17.4, 38.0]%
Expectancy (raw)	+0.35 R /trade	boot95 [-0.10, +0.83], $p=0.15$
Random-direction null	-0.10	[-0.37, +0.18], $p(\text{eng} \geq \text{rand})=0.00$
Anti (flipped) portfolio	-0.54	beats anti, but see drift control
Grade A / Grade B	26% win, +0.40 R ($n=19$)	27% win, +0.33 R ($n=49$)
A beats B? / calibrated?	No (A~B, $n=19$)	No

Drift control (the decisive test). Raw expectancy is mildly positive and beats the direction-randomizing null ($p=0.00$) — but that null randomizes side, not market drift, so a mostly-long book in a rising market beats it by construction. Decomposing by side and benchmarking against simply holding the market removes the illusion:

Benchmark	Mean R	Reading
Long setups	$n=46$, +0.89 R	all the profit is here
Short setups	$n=22$, -0.79 R	they lose
Force every setup long	+0.63 R	beats the engine's +0.35 R
Buy-and-hold from entry	+6.51 R	the actual drift
Engine alpha vs buy-and-hold	-6.16 R	boot95 [-7.97, -4.46]

Reading. The engine's positive expectancy is market drift, not wave skill: profit comes only from long setups, the short calls lose, and going long on every signal (+0.63 R) beats the engine's selective book (+0.35 R) — its directional choices subtract value. Against buy-and-hold the alpha is significantly negative (−6.16 R). So the engine is a strictly worse way to be long than holding the market, and its bootstrap expectancy interval spans zero regardless. Grade and confidence do not track outcomes ($A \approx B$; buckets non-monotonic). Net: no detectable wave-counting edge or directional skill on this sample; any latent signal would have to come from the weighting layer — tested next.

5. Robustness: can tuning or a learned weigher create an edge?

5.1 The pivot-sensitivity switch. The engine's highest-leverage free parameter is the pivot reversal threshold. We replaced the fixed-percentage threshold with a structural one — $\text{threshold} = \ln(1 + k \cdot \text{ATR}/\text{close})$, measured on each timeframe's own bars — so a single coefficient k adapts sensitivity to each instrument's volatility. We then asked whether tuning k manufactures an edge, scoring direction against a geometry-matched random-side null. Selecting k by looking at a window makes that window glow: $k=5$ beat its null at $p \approx 0.03$ and even replicated on a time-held-out split ($p \approx 0.02$). But the decisive test holds out both the parameter and the instruments — one random half of the tickers chose k (it picked $k=4$), the other half was scored once:

Held-out test	n	Expectancy vs random-side null (R)
$k=5$, selected on this window	28	obs +0.640 / null +0.041 ($p=0.029$)
$k=5$, time-held-out (pre-2015)	52	obs +0.777 / null +0.231 ($p=0.018$)
$k=4$, parameter + names held out	20	obs +0.072 / null +0.153 ($p=0.591$)

On the fully held-out half the observed expectancy falls below the geometry null (negative gap, $p=0.59$) — no edge. The 'best' k is unstable ($k=4$ on one half, $k=5$ on the panel holdout) and the obs-vs-null gap collapses from +0.60/+0.55 to -0.08 once selection is forbidden. This reproduces, inside our own pipeline, the central finding of [4]: apparent profitability that survives a single hold-out can still vanish once selection across configurations is accounted for.

5.2 A learned weigher over the same features. We built a supervised table (one row per candidate scenario: the anonymized structural features of §3.2 plus its implied direction; label = whether that direction matched the sign of the forward 252-day return), trained a histogram gradient-boosted tree (depth 3, isotonic-calibrated) on one random half of the instruments, and evaluated once on the other half.

Held-out predictor	AUC	Reading
GBT, all features	0.674	apparently predictive
Implied-direction sign alone	0.675	explains all of it (drift)
GBT, structure only (no direction)	0.529	wave features ≈ chance
GBT, structure only, long setups	0.467	no separation within a side

Held-out AUC of 0.674 is not wave skill: the identical AUC (0.675) comes from the implied-direction sign alone, because over 252-day windows the panel drifts up ($P(\text{correct} \mid \text{long})=0.70$ vs 0.35 for short). Remove direction and wave structure alone gives AUC 0.529 (0.467 within long setups) — chance (label-shuffled null 0.502). The structural features carry essentially no out-of-sample directional information beyond drift, so a learned weigher over them can only follow drift — which §4's benchmark already shows loses to buy-and-hold. The weigher's ceiling is feature information, not model class.

6. LLM scenario weigher — pre-registration & controls

A drop-in weigher (local model via Ollama) re-weights the engine's existing candidate scenarios (and may change lead direction and confidence), leaving pivots/rules/structure/setup untouched. Controls: determinism (temperature 0 + response cache); no leakage (model sees only anonymized relative features — structure, degree, leg proportions, Fibonacci fit, price position); identical universe, tester, horizon. Pre-registered success: expectancy CI clearing zero, grade $A > B$, monotonic confidence-win relationship, and the engine escaping the drift benchmark. A null result localizes the bottleneck in wave-counting rather than weighting.

7. Results — LLM weigher

Metric	Deterministic	LLM (qwen2.5:14b-instruct)
Setups resolved	68	70
Win rate	26.5%	25.7%
Expectancy	+0.35 R [-0.10, +0.83]	+0.31 R boot95 [-0.15, +0.80], $p=0.18$
Grade A / B	26% win, +0.40 R (n=19) / 27% win, +0.33 R (n=40), +0.08 R (n=26) / 30% win, +0.45 R (n=44)	
A beats B? / calibrated?	No (A~B, n=19) / No	No (A<B) / No

The LLM weigher did not improve the engine: expectancy edged down (+0.31R vs +0.35R, CI still spanning zero) and its confidence signal degraded — grade-A (high-confidence) setups returned +0.08R against +0.45R for grade B, and

its top confidence bucket underperformed its lower one. Seeing the same anonymized structural features that carried no out-of-sample directional signal for the gradient-boosted tree (§5.2), it reshuffled confidence on noise rather than adding information. This is the pre-registered null outcome, and it localizes the bottleneck in the wave-count/features, not the weighting model.

8. Toward greater statistical power (future work)

The dominant limitation is sample size (68 setups). Concrete steps: (a) monthly cadence on all 50 instruments and a wider universe — hundreds of setups, tightening every interval; (b) block bootstrap by instrument to respect within-name clustering; (c) explicit benchmarks — buy-and-hold (implemented; the engine loses to it) and a volatility-matched random-entry control; (d) a formal calibration curve with Brier score / expected calibration error once n permits; (e) White's Reality Check or Hansen's SPA when comparing multiple engine/weigher configurations, to bound data-snooping [4]; (f) transaction-cost and slippage sensitivity. Pre-registration (done for the LLM arm) constrains researcher degrees of freedom.

9. Limitations

Underpowered sample; annual cadence misses short-lived setups; one engine family with un-tuned secondary parameters (beam width, R/R floor, horizon scaling); frictionless, point-in-time; a test of one mechanization of EW, not discretionary practice. Research artifact, not trading advice.

10. Conclusion

A reproducible, pre-registered harness gives a precise answer on this universe: the mechanical engine has no statistically detectable out-of-sample edge and no wave-counting skill. Its mildly positive raw expectancy is market drift — long setups profit, short setups lose, an all-long book beats the engine's selective one, and the engine underperforms buy-and-hold by a significant margin. Neither tuning the most powerful parameter (an apparent edge that vanished once parameter and instruments were both held out) nor a learned weigher (whose skill was drift, not structure) nor a local-LLM weigher (which slightly worsened calibration) changed this. The bottleneck is the information in the wave-count features, not the weighting model — reproducing, inside our own pipeline, the data-snooping and no-edge cautions of the technical-analysis literature [3,4].

References

- [1] Frost, A. J. & Prechter, R. R. *Elliott Wave Principle: Key to Market Behavior*. New Classics Library (1978; 10th ed. 2005).
- [2] EW subjectivity/critique summarized in [3,4].
- [3] Lo, A. W., Mamaysky, H. & Wang, J. (2000). Foundations of Technical Analysis. *Journal of Finance* 55(4), 1705–1765.
- [4] Sullivan, R., Timmermann, A. & White, H. (1999). Data-Snooping, Technical Trading Rule Performance, and the Bootstrap. *Journal of Finance* 54(5), 1647–1691.
- [5] Brock, W., Lakonishok, J. & LeBaron, B. (1992). Simple Technical Trading Rules and the Stochastic Properties of Stock Returns. *Journal of Finance* 47(5), 1731–1764.
- [6] D'Angelo, E. & Grimaldi, G. (2017). The Effectiveness of the Elliott Waves Theory to Forecast Financial Markets: Evidence from the Currency Market. *International Business Research* 10(6), 1–18.
- [7] Chendroyaperumal, C. & Karthikeyan, B. Empirical Verification of Elliott Wave Theory in the Indian Stock Market. SSRN working paper 1887789.
- [8] Exploring the Elliott Wave Principle to interpret metal commodity price cycles. *Resources Policy* (Elsevier), 2018.

Artifacts: engine ewt/; testers tester.py, aggregate_tester.py (Wilson/bootstrap/null); disjoint-selection sweep_sensitivity.py (--pivot-mode atr); weigher ewt/weigh/; runners batch_universe.py, weighed_walkforward.py; protocol EXPERIMENT.md, RUN_LLM_ARM.md.